

ripples from the power supply and regulator IC LM7809 (IC1) provides regulated 9V DC to the emitter of pnp transistor T1 (TIP127/BD140) and pin 7 of op-amp IC CA3140 (IC2), which is configured in comparator mode.

The reference voltage of 6.3V at pin 2 of IC2 is obtained through the combination of resistor R7 (1-kilo-ohm) and

zener diode ZD1 (6.3V). The comparator controls charging of the battery. Pin 3 of IC2 is connected to the positive terminal of the battery to be charged through resistor R5. When the battery is fully charged, it stops charging and the green LED (LED2) glows to indicate the full-charge status.

When the battery voltage is low,

diode D1 (1N4007) forward-biases and the battery connects (through resistor R3) to the collector of T1 for charging (indicated by the glowing of red LED1). Three high-wattage white LEDs (LED3 through LED5), such as KLHP3433 from Kvality Photonics, are used for lighting. These are switched on using switch S3. ●

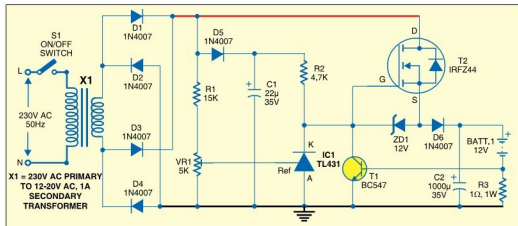
Versatile Battery Charger

T.A. BABU

This transformer-based battery charger circuit can be used to charge any battery, provided the transformer rating is higher than the battery rating. Normally, a linear regulator IC is used for charging. But this charger circuit uses a MOSFET switch, which effectively draws current from the transformer when its voltage is close to the desired battery charging voltage.

The full-wave bridge rectifier rectifies the transformer output and the combination of diode D5 and capacitor C1 provides gate bias voltage to the MOSFET (T2). Zener diode ZD1 maintains the gate voltage at a maximum of 12V relative to the source. The rectified AC potential is also divided by the combination of resistor R1 and 5-kilo-ohm preset VR1 to set the charging voltage.

When the rectified AC voltage is below the threshold of IC1 (TL431), the



MOSFET conducts. Filter capacitor C2 at its output charges through diode D6. As the rectified voltage decreases from peak to 0V, IC1 turns off at approximately 13.6V. Transistor T1 conducts to provide another pulse current to charge capacitor C2. Thus power drain from the transformer occurs in short pulses and the battery charges between the pulses.

To limit the charging current to the battery, the current through the battery is sensed by resistor R3. Once the voltage drop across R3 reaches the base-emitter threshold of transistor T1, it conducts to turn on IC1, which, in turn, switches off MOSFET T2. This process continues, limiting the charging current to the battery. ●

DC Motor Control Using PWM

STUTEE SAXENA

Small DC motors are efficiently controlled using pulse-width modulation (PWM) method. The circuit described here is built around

an LM324 low-power quad-operational amplifier. Of the four op-amps (operational amplifiers) available in this IC, two are used for triangular wave generator and one for comparator.

Op-amp N2 generates a 1.6kHz

square wave, while op-amp N1 is configured as an integrator. The square wave output of N2 at its pin 14 is fed to the inverting input (pin 2) of N1

